

Remember that each digit $c_{i,j}$ is an integer between 0 and 9. Let's now look at the numbers $c_{i,i}$ on the *diagonal*. Consider a real number between 0 and 1 defined as:

$$r = 0. d_1 d_2 d_3 d_4 d_5 \dots \quad \text{where } d_i \neq c_{i,i} \quad \forall i \in \mathbb{N}_+$$

(and also d_i shouldn't be 0 or 9: *can you think why?*). Then

$$\begin{array}{ll} r \neq r_1 & \text{since the } \textit{first} \text{ digit in their expansions is different} \\ r \neq r_2 & \text{since the } \textit{second} \text{ digit in their expansions is different} \\ r \neq r_3 & \text{since the } \textit{third} \text{ digit in their expansions is different} \\ \vdots & \\ r \neq r_n & \text{since the } \textit{nth} \text{ digit in their expansions is different} \\ \vdots & \end{array}$$

Hence r isn't equal to any of the reals that have been enumerated. However, that is a contradiction to the assumption that all reals in $(0,1)$ have been enumerated. $\square$

This method of proof is called a **diagonal argument**, and has become an important technique for proving various results in the years since Cantor introduced it.

2.4 Cartesian product

Ordered pairs and Cartesian product

Let X and Y be two nonempty sets. Then we define an **ordered pair** to be

$$(x, y)$$

where $x \in X$ and $y \in Y$. The set of all ordered pairs from X and Y is called the **Cartesian product of X and Y** and is denoted $X \times Y$:

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}.$$

When $X = Y$ we often write X^2 rather than $X \times X$.

The order of elements in an ordered pair is important: the first *component* belongs to X and the second *component* belongs to Y . Thus, the ordered pair (x, y) is *fundamentally different* from the set $\{x, y\}$.

Example 2.6: 1. One of the most important examples of a Cartesian product, which we will often use, is that of the **plane** $\mathbb{R}^2$. This is the set of all ordered pairs of real numbers:

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

This is called the **Cartesian plane**.

2. A practical example is that of latitude and longitude: we can describe points on Earth in terms of their angle north/south of the equator, and east/west relative to a chosen Prime Meridian (a big circle connecting the North and South poles). Normally this is chosen to be the Greenwich Meridian, passing through the Royal Observatory in Greenwich, England (this choice dates back to 1884). In this system, the Colosseum, for instance, is located at $(41.8902^\circ\text{N}, 12.4922^\circ\text{E})$.

An ordered *pair* need not be limited to two components. One can have n elements from n sets, where $n \in \mathbb{N}$:

$$(x_1, x_2, \dots, x_n), \quad x_i \in X_i, \quad i = 1, \dots, n.$$

This is called an **n -tuple**. Then the Cartesian product involves the sets $X_1, \dots, X_n$:

$$X_1 \times X_2 \times \dots \times X_n.$$

If $X_i = X$ for all $i = 1, \dots, n$, then we simply write $X \times X \times \dots \times X = X^n$.

2.5 Relations in the Cartesian plane

In the Cartesian plane (we will simply call it *the plane*) $\mathbb{R}^2$ we typically denote the first coordinate by x and the second by y . We can describe subsets of $\mathbb{R}^2$ by equations and inequalities involving x and y . This is best understood using some examples:

Example 2.7: Sets described by equations:

1. The equation $y = 0$ describes the x -axis.
2. The equation $x = 0$ describes the y -axis.
3. The equation $x = y$ describes the line through the origin with slope 1.
4. The equation $x^2 + y^2 = 1$ describes the circle of radius 1 around the origin.

Example 2.8: Sets described by inequalities:

1. The inequality $y < 0$ describes the *lower half plane* excluding the x axis.
2. The inequality $x \geq 0$ describes the *right half plane* including the y axis.
3. The inequality $x < y$ describes the half plane to the left of the line through the origin with slope 1.
4. The inequality $x^2 + y^2 \leq 1$ describes the interior of the circle of radius 1 around the origin, including its boundary.

2.6 Factorials and binomial coefficients

Here we briefly discuss notions that arise in mathematical fields such as *Discrete Mathematics*, *Combinatorics* and *Probability*. The most basic problem is as follows. Consider the set X of n students

$$X = \underbrace{\{\text{Andrea, Marta, Jim, Victoria, } \dots, \text{Caterina}\}}_{n \text{ students}}$$

As this is a *set*, there is no importance for the ordering of the students. We could express X also as

$$X = \{\text{Caterina, Marta, Victoria, Andrea, } \dots, \text{Jim}\}$$

However, the elements of X (the n students) are all distinct, and we often care about their ordering. We can ask two natural questions:

1. In how many ways can we order the n students? Above we have seen two examples of how to order (or **permute**) the students. There is a simple formula that gives us the number of *permutations*:

- We start by choosing the first student. There are n students in total, hence we have n students to choose from.
- Next, we want to choose the second student to appear in our ordering. We have already chosen one student, so there are only $n - 1$ students left to choose from.
- For the third student we have $n - 2$ students to choose from.
- And so on....
- For the $(n - 1)$ st student we have two students to choose from.
- For the n th student we no longer have a choice.

Hence, we find that the number of possible permutations of n elements is

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1.$$

n factorial

Since this is an important formula, it has its own special symbol:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1$$

which is called n **factorial**. We also define

$$0! = 1.$$

2. In how many ways can we choose k students of the n students? Repeating the same argument as above, we have

- n options for the first student,
- $n - 1$ options for the second student,
- $n - 2$ options for the third student,
- and so on....
- $n - (k - 1) = n - k + 1$ options for the k th student.

So we get

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - k + 2) \cdot (n - k + 1).$$

We observe that this expression can be simplified:

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - k + 1) = \frac{n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1}{(n - k) \cdot (n - k - 1) \cdot (n - k - 2) \cdot \dots \cdot 2 \cdot 1} = \frac{n!}{(n - k)!}$$

Now we make an important observation: the ordering of the k students *does not matter for us*. So, for example if $k = 2$, there is no difference for us between $\{\text{Jim, Victoria}\}$ and $\{\text{Victoria, Jim}\}$. So we need to eliminate such repetitions. But these repetitions are precisely the number of possible permutations of k elements, which we have seen: it is $k!$. Hence we need to divide the above formula by this number.

n choose k

The number of possible ways to choose k elements from a total of n elements (where $0 \leq k \leq n$) is called n choose k and is denoted

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}$$

This is also called the **binomial coefficient**.

It turns out that these coefficient satisfy certain recursive relations (that is, one can compute $\binom{n}{k}$ from knowledge of $\binom{n-1}{j}$ for all $0 \leq j \leq n - 1$). The simple way to visualize this is through *Pascal's triangle*, whose first eight lines look as follows:

$$\begin{array}{ccccccc} & & & & 1 & & & \\ & & & & 1 & & 1 & \\ & & & 1 & & 2 & & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\ 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\ 1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1 \end{array}$$

If we let n denote the line number (starting from 0) and k denote the position of a given number (also, starting from 0) within a line, then the number appearing in the triangle is precisely $\binom{n}{k}$. Hence, for instance,

$$1 = \binom{0}{0} = \binom{n}{0} = \binom{n}{n}$$

for all $n \in \mathbb{N}$, and

$$n = \binom{n}{1} = \binom{n}{n-1}$$

for all $n \in \mathbb{N}_+$. Some other specific examples are

$$6 = \binom{4}{2}, \quad 10 = \binom{5}{2} = \binom{5}{3}, \quad 35 = \binom{7}{3} = \binom{7}{4}.$$

Exercise 2.1: Can you see the pattern?

Newton's binomial formula

For any $n \in \mathbb{N}$ and $a, b \in \mathbb{R}$, there holds **Newton's binomial formula**:

$$(a + b)^n = a^n + na^{n-1}b + \cdots + \binom{n}{k}a^{n-k}b^k + \cdots + nab^{n-1} + b^n$$

Exercise 2.2: Prove Newton's binomial formula.

Summation notation

Let $n, N \in \mathbb{Z}$ with $N \geq n$. Let $c_n, \dots, c_N \in \mathbb{R}$. Then the expression

$$\sum_{k=n}^N c_k$$

is a concise way to write the sum

$$c_n + \dots + c_N.$$

The integers n and N are known as the **lower** and **upper limits of summation**, respectively. The **subscripts** $n, \dots, N$ are called the **indices**. The numbers c_k ($k = n, \dots, N$) are called the **summands**. The summation can also be denoted:

$$\sum_{k=n}^N = \sum_{k \in \{n, \dots, N\}}$$

indicating that the summation is over the set of integer indices $\{n, \dots, N\}$. The symbol k is known as the **summation index** and it is a **dummy variable**: this means that it can be replaced by any other symbol without changing the meaning of the expression:

$$\sum_{k=n}^N c_k = \sum_{j=n}^N c_j = \sum_{m=n}^N c_m = \sum_{\star=n}^N c_{\star} = \sum_{\clubsuit=n}^N c_{\clubsuit}$$

The summation notation allows us to simplify the expression for Newton's binomial formula:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$